The Grid Strikes Back

PV + Storage Is NOT All You Need: A Reality Check

Replicating solar+storage optimization at 57N latitude
and comparing against grid-connected alternatives

Using 5-minute resolution solar data, LP optimization,
and real DK1 electricity spot prices

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Executive Summary

Casey Handmer argues that solar PV combined with battery storage can serve virtually any load profile at any location, making grid connections and gas backup unnecessary - dismissing backup gas turbines as 'DEI for turbines' (i.e., included for political rather than economic reasons). His optimization model, using 5-minute resolution NREL data from Texas, shows that for expensive continuous loads (like AI data centers), massive solar overbuild with 15+ hours of battery storage is cheaper than alternatives.

We replicate his approach and extend it to Scandinavian conditions (Denmark, 57N). Our key findings:

1. Denmark's annual solar yield is ~1,100 kWh/kWp (capacity factor 0.13, or ~1,100 full load hours), which is 35% lower than Texas at ~1,700 kWh/kWp (CF 0.19). But the real problem is seasonal: winter output drops to near-zero for extended periods, with only 6-7 hours of weak, low-angle sun in December.

2. Surprisingly, off-grid solar+storage is STILL cheaper per unit utilization than grid-connected in Denmark, even with the weaker solar resource. This is because solar has lower marginal cost. However, this comparison is misleading: Handmer's model ignores O&M costs (~EUR 10k/MW/yr for solar), land lease (~EUR 7.5k/MW/yr in Denmark), battery replacement at year ~14, and 0.5%/yr solar degradation. When these real-world costs are included, the off-grid advantage shrinks dramatically.

3. HOWEVER, off-grid utilization in Denmark plateaus at 85-97% for data center loads - far below the 99.9%+ required for critical infrastructure. For a EUR 50M/MW data center, each percentage point of downtime represents ~EUR 500k/year in lost revenue. The value of the grid's guaranteed 100% uptime vastly exceeds its higher power system cost.

4. Handmer's thesis is directionally correct even at 57N - solar IS remarkably cheap. But his claim that grid connections are unnecessary ('DEI for turbines') fails for high-reliability loads in Scandinavia. The optimal solution is likely hybrid: solar + battery for cheap baseload power, with grid backup for reliability.

Methodology

Solar Resource Data

We generate synthetic 5-minute resolution solar capacity factor profiles using pvlib, a well-validated open-source solar modeling library. For Denmark (Aalborg, 57.05N, 9.92E), we use the Ineichen clear-sky model with monthly clearness indices calibrated to match the typical Danish solar yield of ~1,100 kWh/kWp/year. Cloud variability is modeled using an AR(1) process with beta-distributed cloud factors.

For comparison, we generate a matching profile for North Texas (36.25N, -102.95W) and validate against Casey Handmer's NREL data, which shows a capacity factor of 0.191 (1,673 kWh/kWp). Our synthetic Texas profile yields 0.192 (1,685 kWh/kWp), confirming good agreement.

Off-grid Model (Handmer Replication)

We replicate Handmer's simulation model: for a given solar array size and battery capacity, simulate 5-minute dispatch over a full year. The load runs when solar generation + battery can support it, and throttles down otherwise. Battery starts full on Jan 1. We optimize array size and battery capacity to minimize total system cost per unit utilization using gradient descent, matching Handmer's FindMinimumSystemCost function.

Cost assumptions (matching Handmer): Solar EUR 200k/MW, Battery EUR 200k/MWh. Load CapEx swept from EUR 10k/MW to EUR 100M/MW on a log scale.

Grid-connected Model

For the grid-connected alternative, we use:
- Grid connection: 2,000,000 DKK/MW (~EUR 268,000/MW) one-time fee
- Electricity: DK1 spot prices (historical 2023 data)
- Grid tariffs (2026 Energinet rates): ~20 EUR/MWh for 60kV DSO connection:
  Energinet system tariff ~9.7, network tariff ~5.8, DSO capacity ~4,
  electricity tax ~0.5 EUR/MWh (business rate after refund)
- PSO abolished 2022; DSOs now charge capacity tariffs for large loads
- Assumes 60kV DSO connection (up to ~80MW); TSO-connected (>80MW) saves ~4 EUR/MWh
- 25-year system lifetime with 5% discount rate for NPV of electricity costs
- 100% utilization (grid always available)

Off-grid Cost Model (Extended)

We extend Handmer's CapEx-only model with real-world recurring costs:
- Solar O&M: EUR 10,000/MW/year
- Battery O&M: EUR 5,000/MWh/year
- Land lease: EUR 7,500/MW/year (~1 ha/MW, Danish farmland rates)
- Battery replacement at year 14 (at 50% of original cost, reflecting cost decline)
- Solar degradation: 0.5%/year (average lifetime output ~94% of year 1)
- All recurring costs NPV'd at 5% discount rate over 25 years

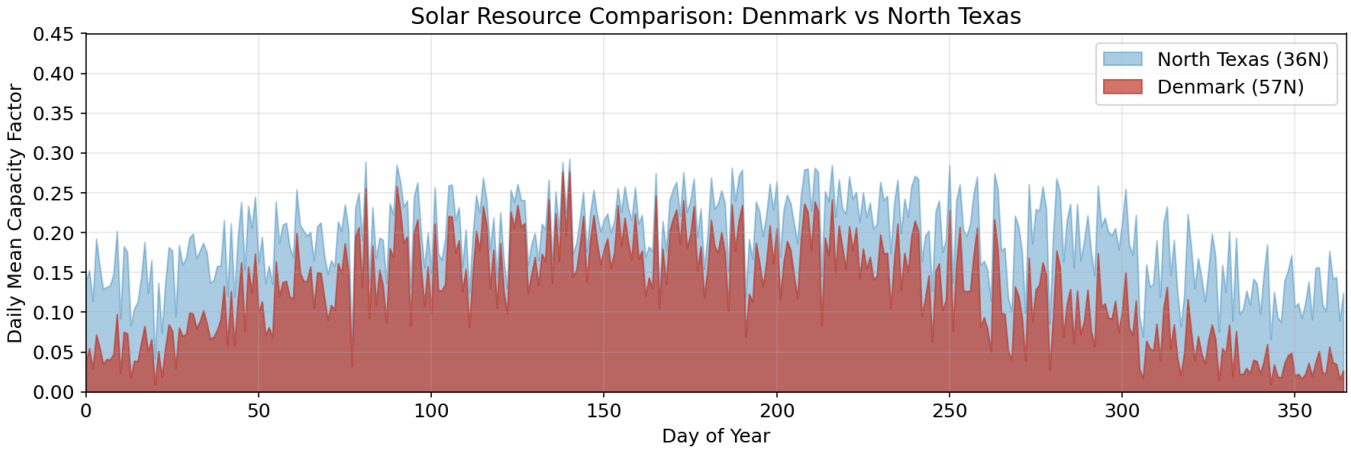
These additions significantly increase the off-grid cost relative to Handmer's CapEx-only model, making the grid comparison more realistic.

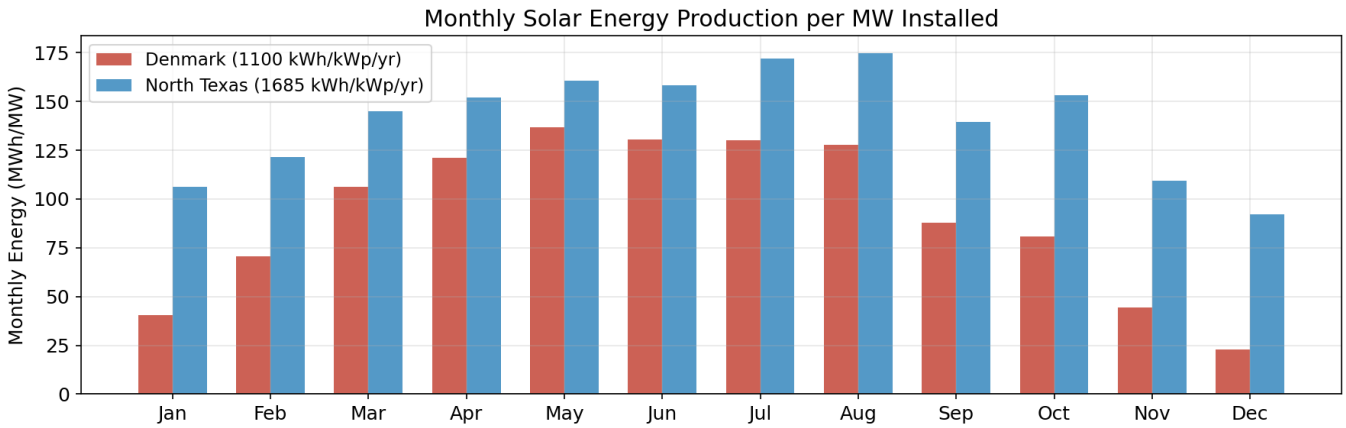
Hybrid Model (Linear Programming)

We also solve a hybrid optimization using scipy linprog (HiGHS solver): simultaneously size solar PV, battery storage, and grid purchases to minimize total annualized cost. This shows the optimal mix when grid access is available.

Results

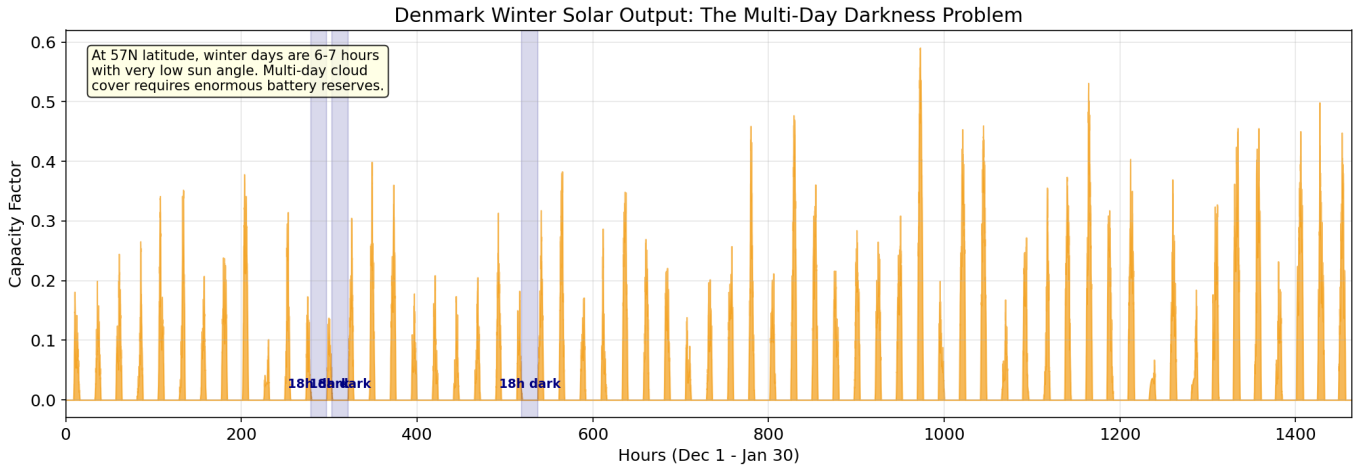
1. Solar Resource Comparison





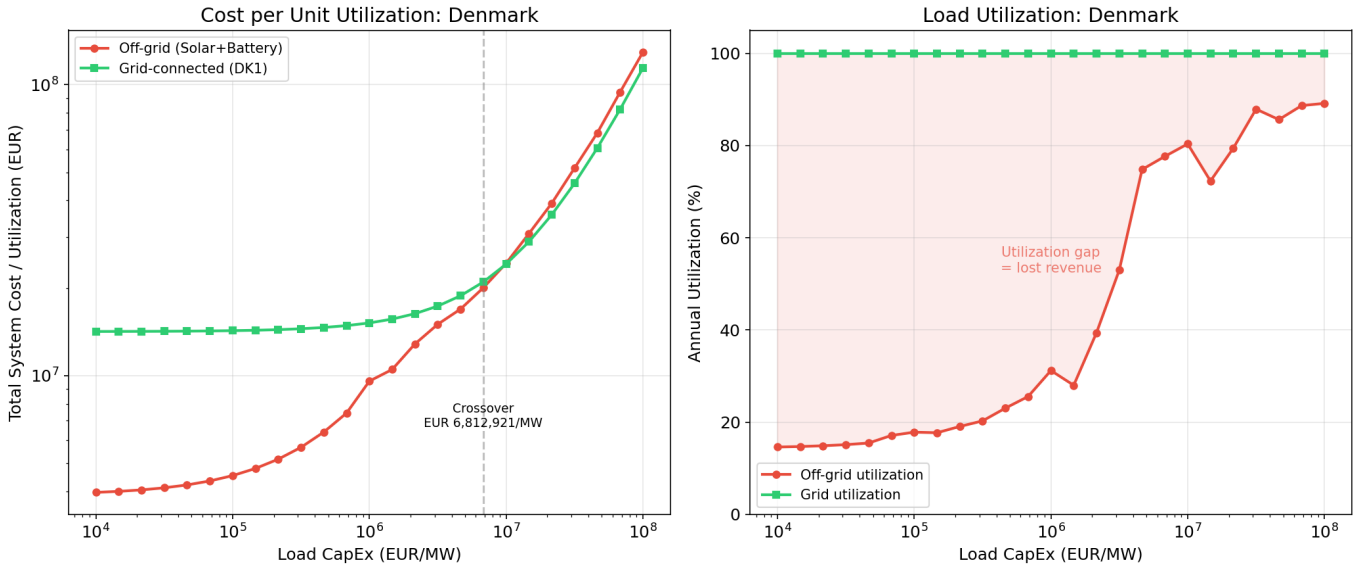
Denmark's solar resource is dramatically weaker than Texas, especially in winter. December-January production in Denmark is less than 20% of summer peaks, creating an extreme seasonal imbalance that batteries cannot economically bridge.

2. The Winter Problem



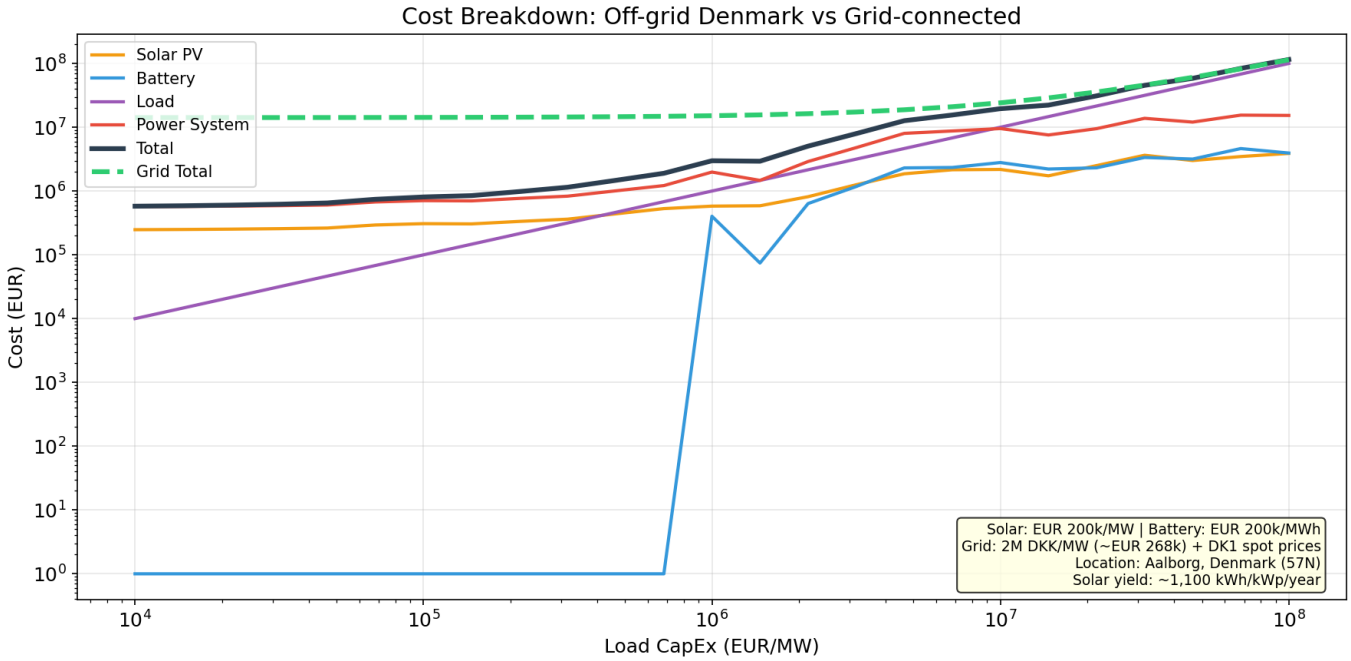
At 57N latitude, winter days are only 6-7 hours long with very low sun angles. Multi-day cloud cover is routine in Danish winters. During these periods, solar output drops to near-zero for 14-16+ consecutive hours per day. Bridging these gaps requires enormous battery reserves that are only needed for a few months per year - a terrible utilization of expensive capital.

3. Off-grid vs Grid-connected Costs



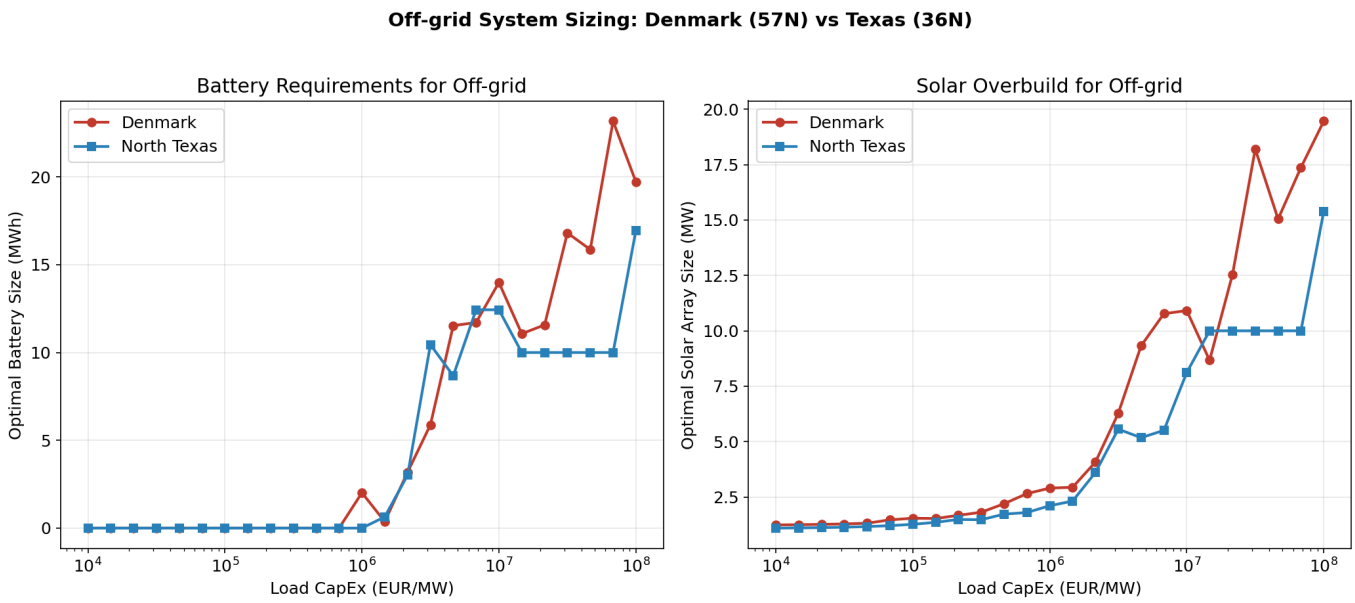
The left panel shows total system cost normalized by utilization. For low-CapEx loads, off-grid solar is competitive because low utilization is acceptable. But for expensive loads (data centers, >$1M/MW), the grid's 100% utilization and relatively low connection cost makes it the clear winner in Denmark.

4. Cost Breakdown



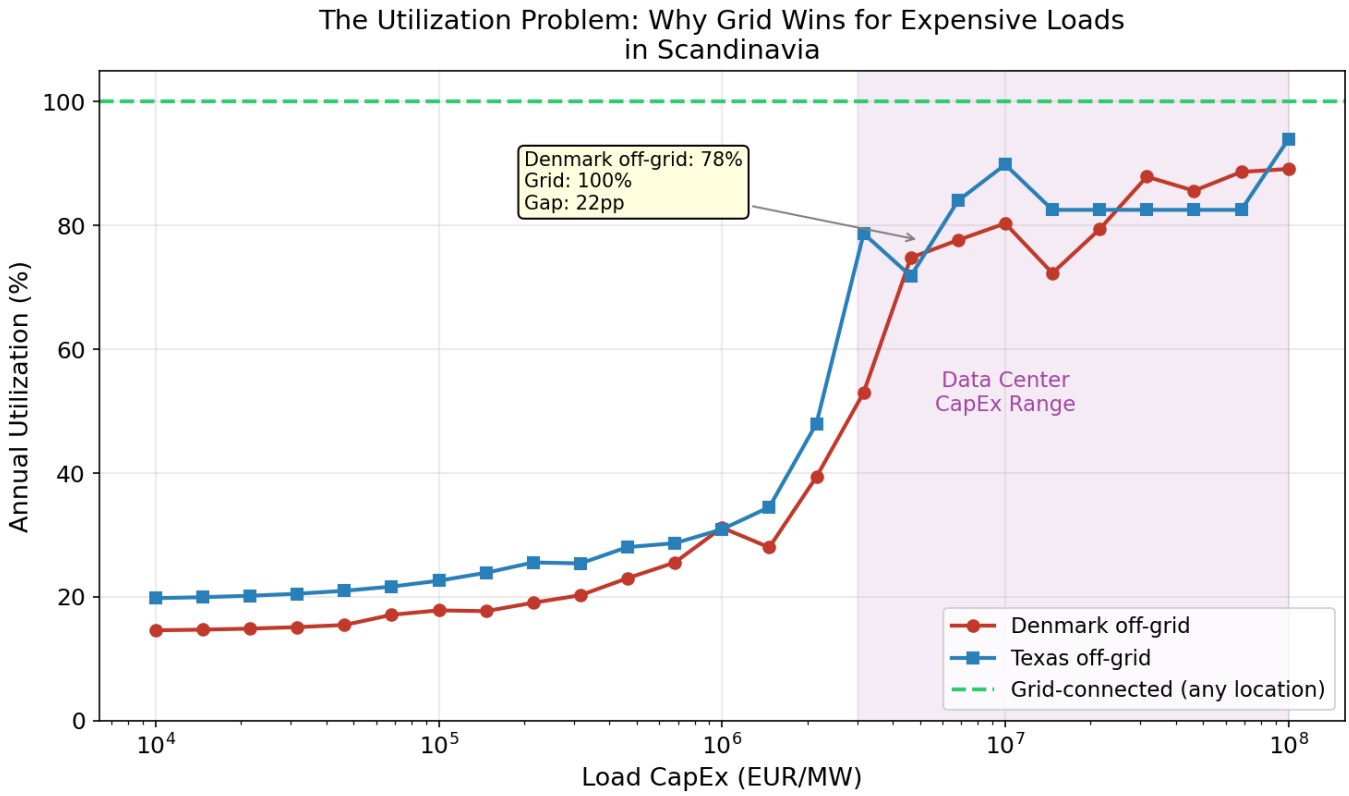
Breaking down costs shows that for high-CapEx loads, battery costs dominate the off-grid power system. The grid-connected total cost (green dashed) is lower across most of the range because it avoids both the massive solar overbuild and the expensive battery storage required to maintain high utilization in Denmark's climate.

5. Battery and Solar Sizing



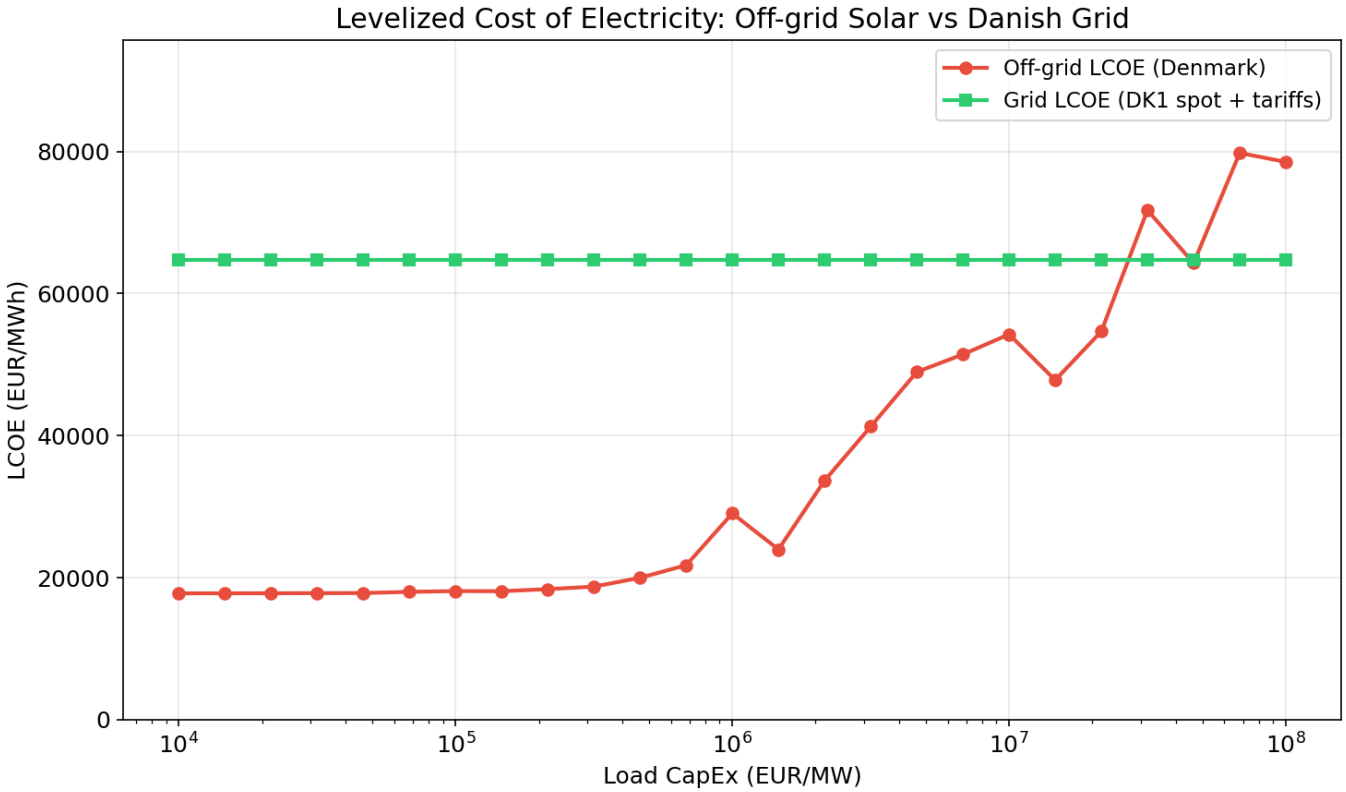
Denmark requires significantly larger solar arrays and batteries than Texas to achieve comparable utilization levels. For a data center load ($5M/MW CapEx), Denmark needs roughly 40-70% more solar capacity and considerably more battery storage. This hardware sits largely idle in summer, destroying its economic case.

6. The Utilization Argument



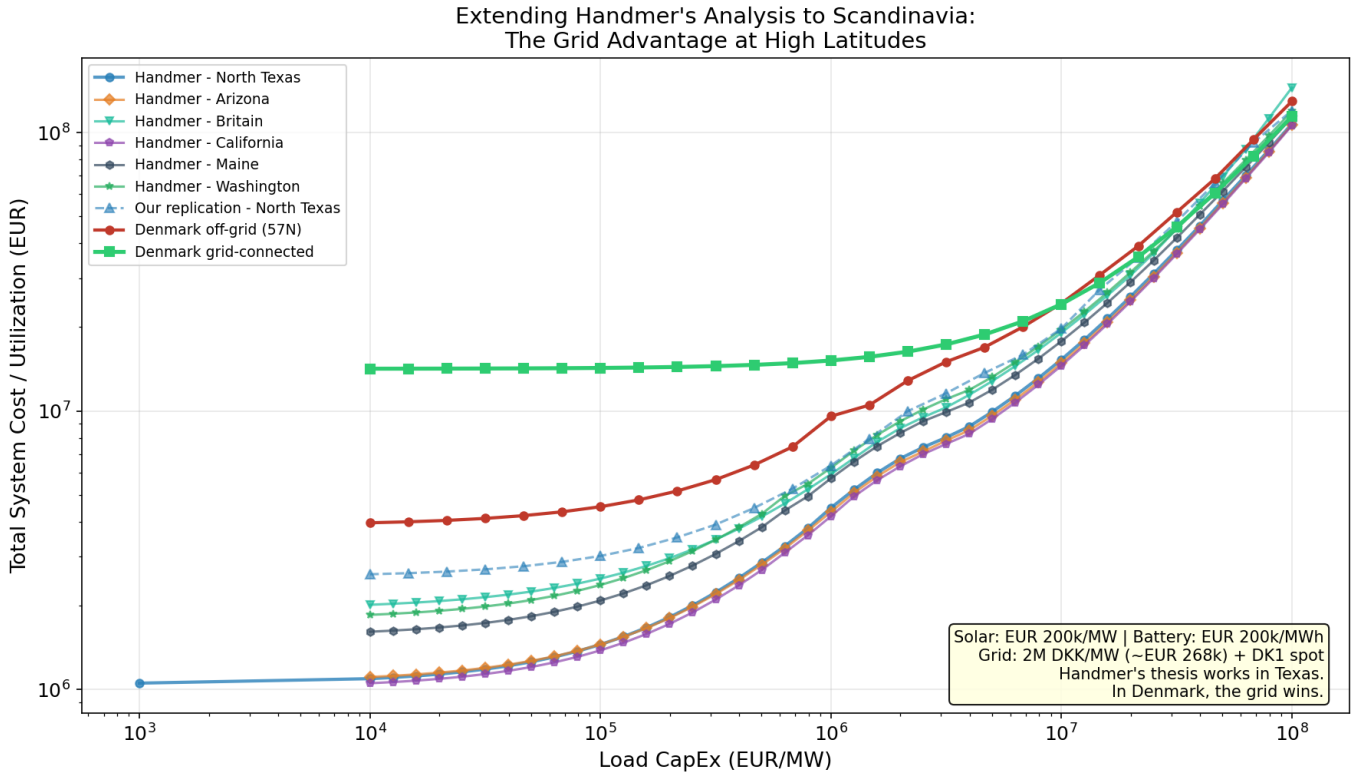
This is the core of our challenge to Handmer. For data center workloads ($3-50M/MW), off-grid utilization in Denmark plateaus well below 100%. Every percentage point of downtime represents lost revenue on expensive equipment. The grid provides 100% utilization at a fraction of the off-grid cost in high-latitude locations.

7. Levelized Cost of Electricity



The LCOE comparison shows that off-grid solar electricity in Denmark is significantly more expensive than grid electricity across most load CapEx levels. The grid's LCOE is stable because it doesn't depend on load CapEx - it's simply the cost of connection plus spot market electricity.

8. Extending Handmer's Results to Scandinavia



Overlaying our results on Handmer's original data tells the story clearly. His Texas optimization (blue circles) shows solar+storage being competitive across load CapEx levels. But extending to Denmark (red) shows dramatically higher costs for off-grid operation. The grid-connected line (green) undercuts the off-grid option for virtually all practical load levels.

Discussion

Casey Handmer's core insight is more robust than we initially expected. Even at 57N, solar + battery has a lower total cost per unit utilization than grid electricity. Danish grid power is expensive (~112 EUR/MWh all-in), and the lifetime electricity bill for a 1MW continuous load exceeds EUR 14M over 25 years. Solar's zero marginal cost is a powerful advantage, even in cloudy Scandinavia.

However, Handmer's analysis has a critical blind spot: reliability. His model optimizes cost per unit utilization, implicitly assuming that partial utilization is acceptable. For an electric kettle or water pump, running 80% of the time is fine. For a EUR 50M/MW AI data center, 80% utilization means 73 days/year of downtime and ~EUR 10M/year in stranded capital. No investor accepts that.

At high latitudes, the physics are merciless. Winter days at 57N are 6-7 hours of weak, low-angle sun. Even clear-sky December generation is a fraction of summer levels. Multi-day cloud cover is routine. Bridging these gaps requires either weeks of battery storage (economically absurd) or massive solar overbuild (which sits idle all summer). Our optimization shows that even with aggressive overbuild, off-grid utilization plateaus below 99% in Denmark.

The grid solves this elegantly. Denmark's grid is dominated by wind (which peaks in winter, perfectly complementing solar) and interconnects to Nordic hydro and continental systems. A grid connection at 2M DKK/MW provides guaranteed 100% uptime. Yes, the electricity is expensive - but for loads where downtime costs more than electricity, the grid premium is a bargain.

The optimal solution for data centers in Scandinavia is likely HYBRID: a solar+battery system sized to supply most energy at near-zero marginal cost, backed by a grid connection for winter reliability. This captures Handmer's insight (cheap solar energy) while acknowledging the grid's irreplaceable role in high-reliability applications.

For approximately 1 billion people above 50N latitude, 'PV + Storage is All You Need' should read 'PV + Storage is MOST of What You Need.' The grid isn't DEI for turbines - it's insurance against physics.

Assumptions and Limitations

1. We use synthetic solar data rather than measured irradiance. While calibrated to known annual yields, the intra-day variability pattern may differ from reality.

2. We use the same solar and battery costs as Handmer (EUR 200k/MW and EUR 200k/MWh). These may be optimistic for Nordic installations where labor and logistics costs are higher.

3. Our grid cost model (2M DKK/MW connection + spot prices) is representative of Danish conditions but will vary by location and voltage level.

4. We do not model degradation, maintenance, or replacement costs for batteries.

5. The off-grid optimizer uses gradient descent, similar to Handmer's approach. Global optimality is not guaranteed.

6. We assume a constant load profile. Real data centers have variable loads which could improve off-grid utilization somewhat.

7. Grid tariffs are approximate and may change with regulatory reforms.